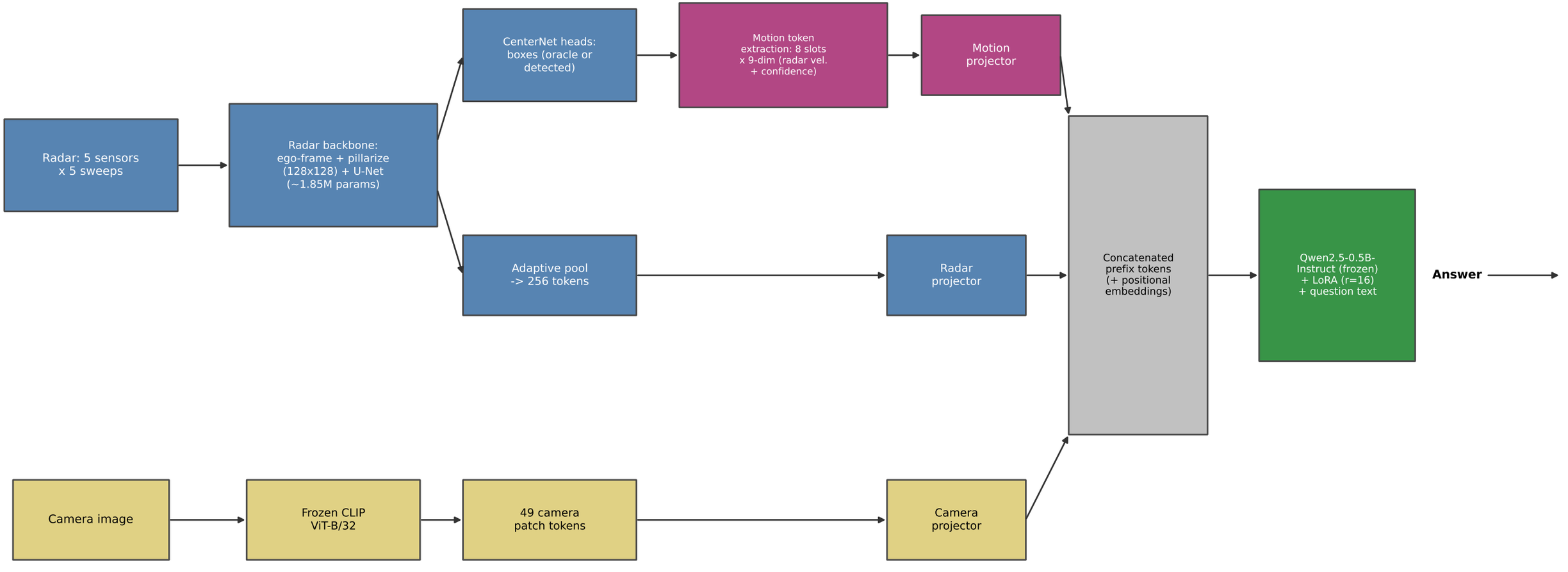
Radar: 5 sensors
x 5 sweeps
Radar backbone:
ego-frame + pillarize
(128x128) + U-Net
(~1.85M params)
CenterNet heads:
boxes (oracle or
detected)
Motion token
extraction: 8 slots
x 9-dim (radar vel.
+ confidence)
Motion
projector
Adaptive pool
-> 256 tokens
Radar
projector
Concatenated
prefix tokens
(+ positional
embeddings)
Qwen2.5-0.5B-
Instruct (frozen)
+ LoRA (r=16)
+ question text
Answer
Camera image
Frozen CLIP
ViT-B/32
49 camera
patch tokens
Camera
projector